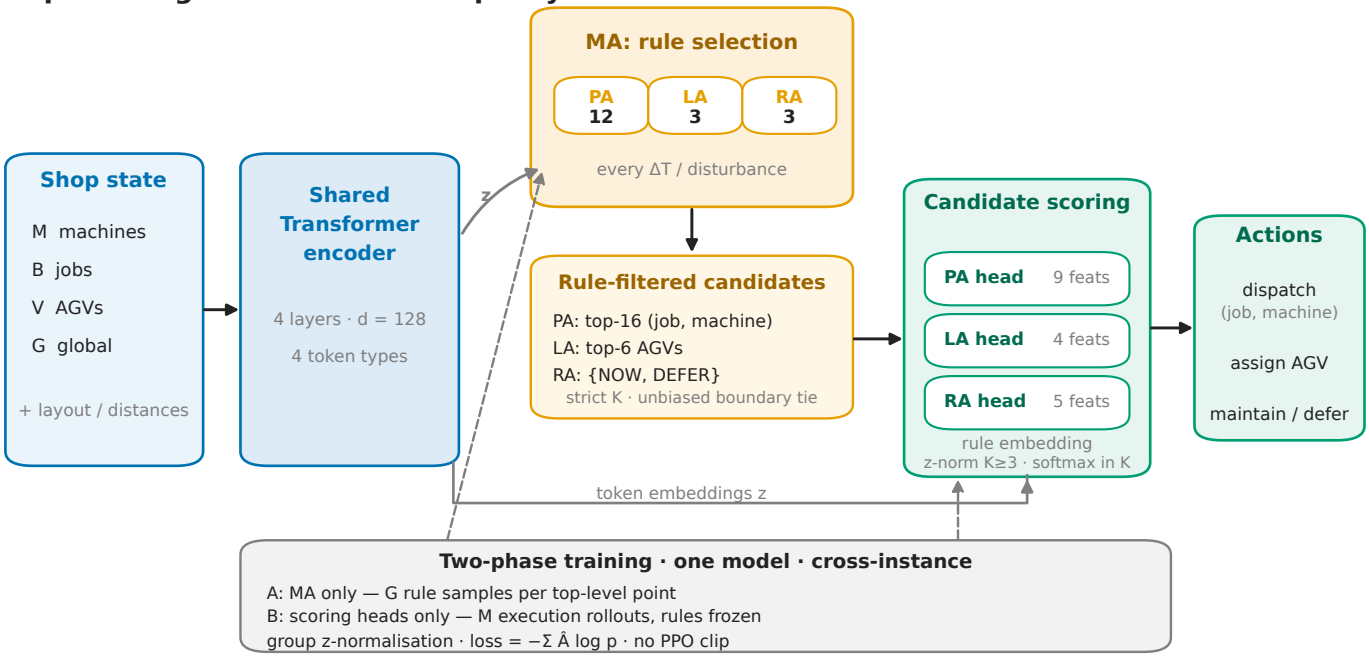

Top-K rule-guided hierarchical policy
MA: rule selection
PA
12
LA
3
RA
3
every ΔT / disturbance
Shop state
M machines
B jobs
V AGVs
G global
+ layout / distances
Shared Transformer encoder
4 layers · d = 128
4 token types
z
Rule-filtered candidates
PA: top-16 (job, machine)
LA: top-6 AGVs
RA: {NOW, DEFER}
strict K · unbiased boundary tie
Candidate scoring
PA head 9 feats
LA head 4 feats
RA head 5 feats
rule embedding
z-norm K≥3 · softmax in K
Actions
dispatch
(job, machine)
assign AGV
maintain / defer
token embeddings z
Two-phase training · one model · cross-instance
A: MA only — G rule samples per top-level point
B: scoring heads only — M execution rollouts, rules frozen
group z-normalisation · loss = −Σ Â log p · no PPO clip